Software Requirements Engineering

Lecture 3: Types of Software Requirements

Review

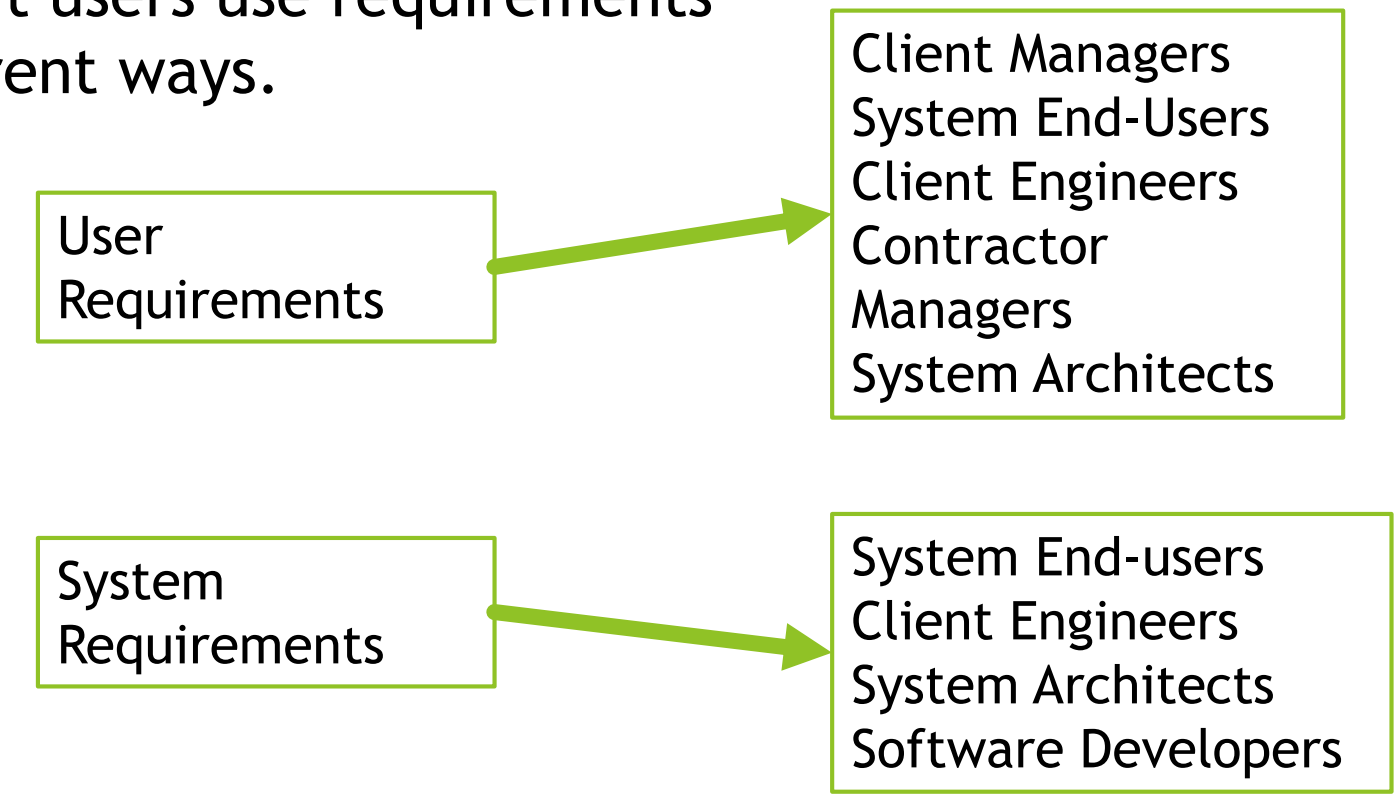
1. Software process models
2. Requirements engineering and software development
3. Software risk management

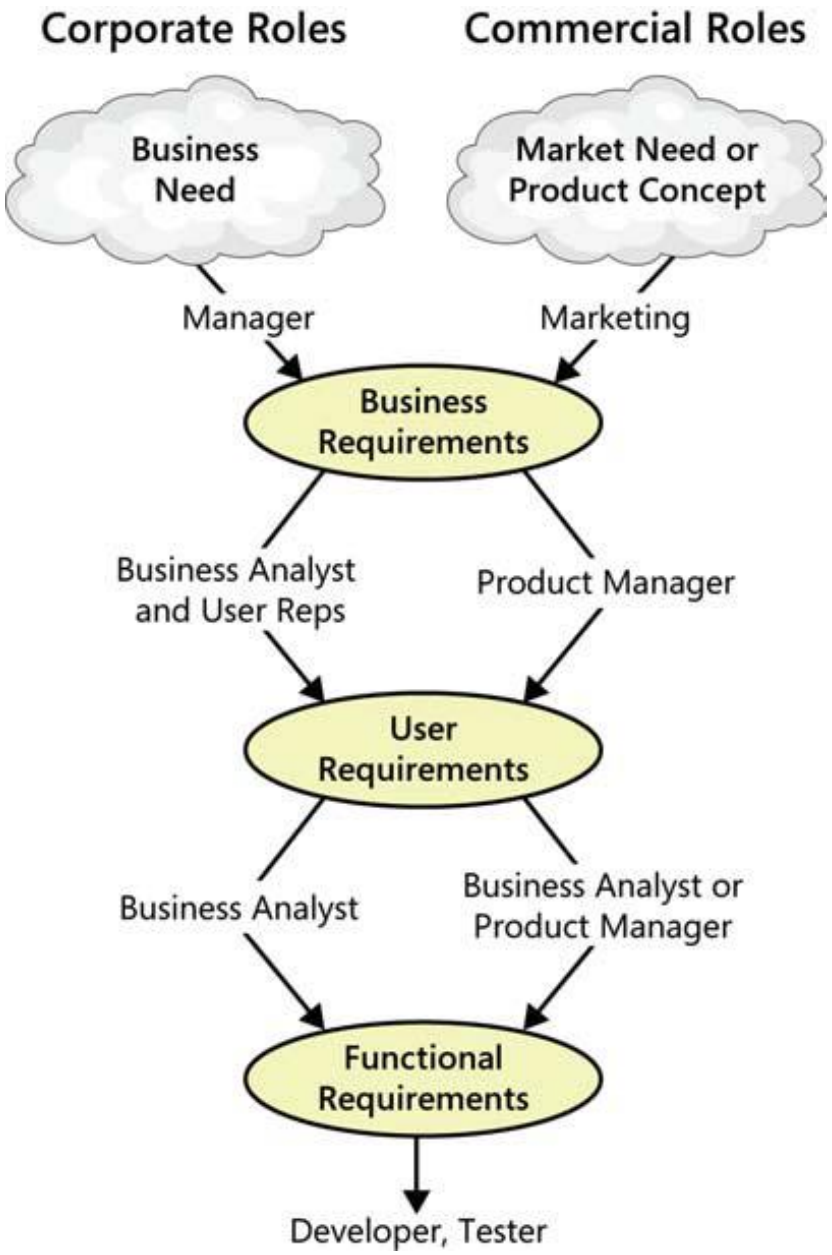
Outline

1. Who is the user?
2. Types and levels of requirements
3. Functional and non-functional requirements

3.1 Levels and Types of Requirements

Different users use requirements in different ways.





An example of how different stakeholders participate in requirements development.

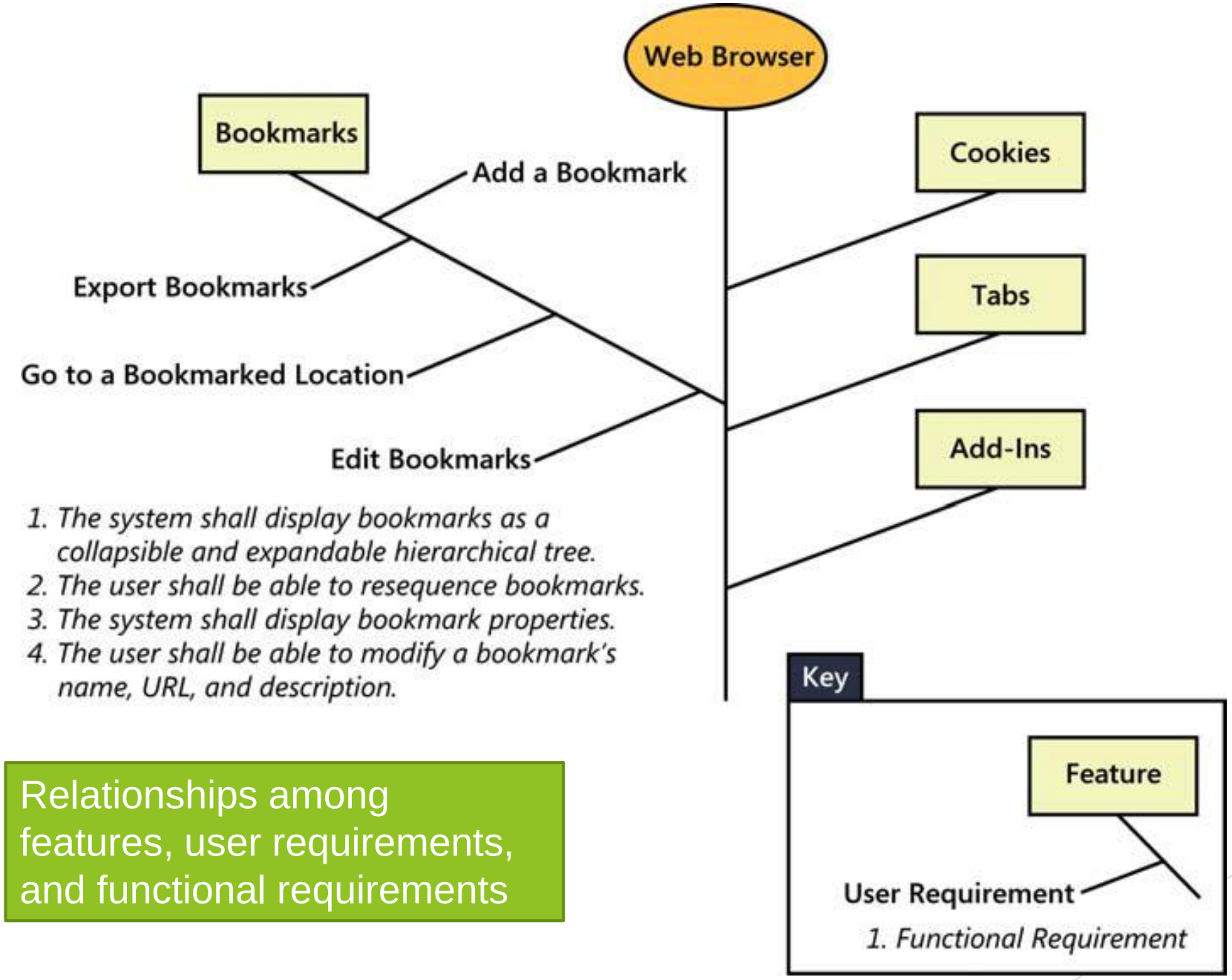
Terms

Who is the customer?

Who is the end user?

- User
  - The one who uses the service provided by the system
  - The direct user of the software system
- Customer
  - The one who pay for the service
  - Usually have the final say of the software requirements
  - Can be the user but does not have to be
- Stakeholder
  - Anyone who benefits in a direct or indirect way from the system
  - Include all the users, customers and the developers
  - They may have different views of the software system

Term	Definition
Business requirement	A high-level business objective of the organization that builds a product or of a customer who procures it.
Business rule	A policy, guideline, standard, or regulation that defines or constrains some aspect of the business. Not a software requirement in itself, but the origin of several types of software requirements.
Constraint	A restriction that is imposed on the choices available to the developer for the design and construction of a product.
External interface requirement	A description of a connection between a software system and a user, another software system, or a hardware device.
Feature	One or more logically related system capabilities that provide value to a user and are described by a set of functional requirements.
Functional requirement	A description of a behavior that a system will exhibit under specific conditions.
Nonfunctional requirement	A description of a property or characteristic that a system must exhibit or a constraint that it must respect.
Quality attribute	A kind of nonfunctional requirement that describes a service or performance characteristic of a product.
System requirement	A top-level requirement for a product that contains multiple subsystems, which could be all software or software and hardware.
User requirement	A goal or task that specific classes of users must be able to perform with a system, or a desired product attribute.

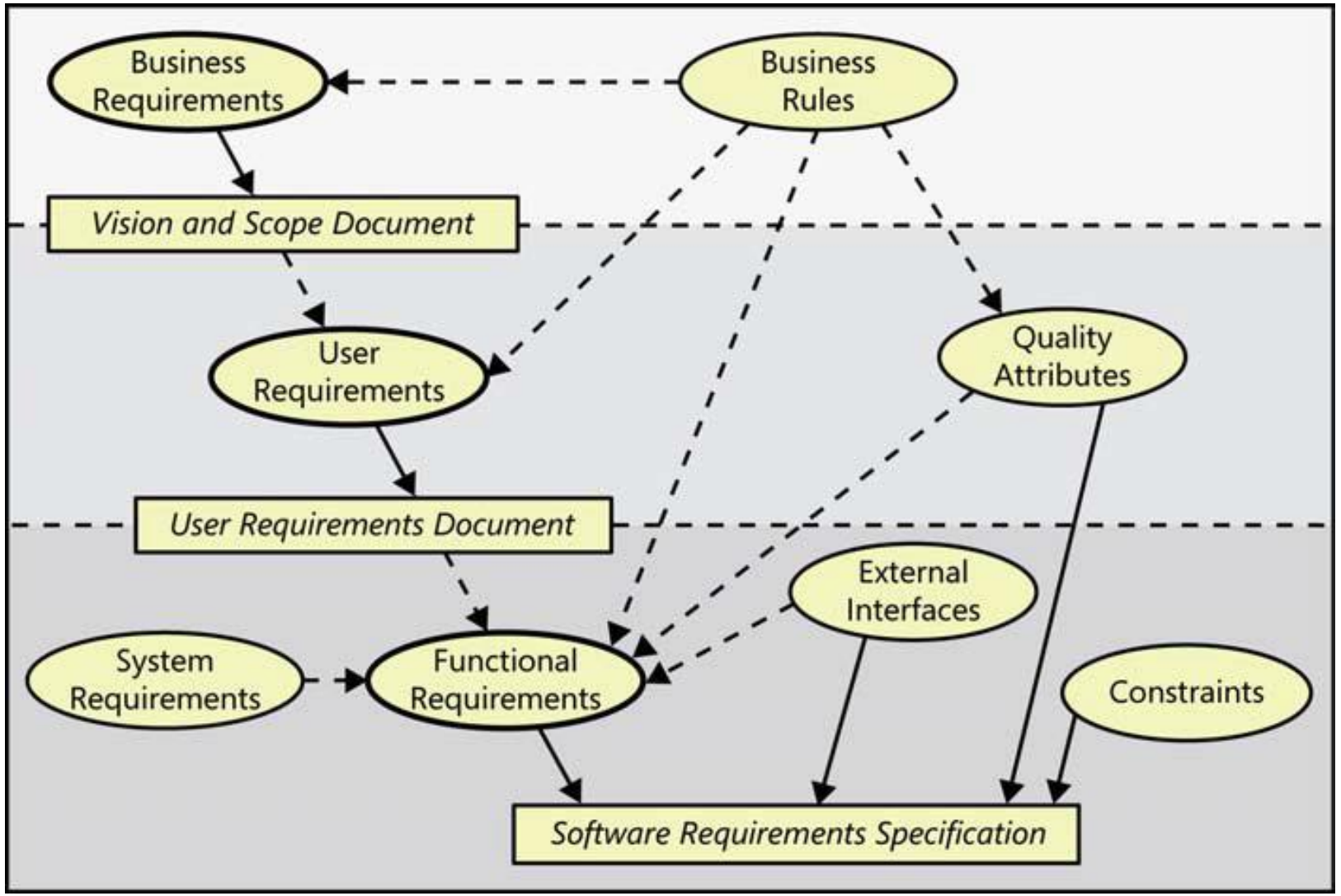


Relationships among features, user requirements, and functional requirements

Different types of requirements

Consider a project to develop the next version of a text editor program

- Business requirement: Increase non-US sales by 25 percent within 6 months.
- User requirement: Select language for spelling checker, Find spelling errors, Add a word to a dictionary
- Functional requirements: Highlighting misspelled words, autocorrect, displaying suggested replacements, and globally replacing misspelled words with corrected words
- Usability requirements: How the software is to be localized for use with specific languages and character sets.



Relationships among several types of requirements information

Types of Requirements

- Functional requirements
  - Statements of services the system should provide, how the system should react to particular inputs and how the system should behave in particular situations.
  - May state what the system should not do.
- Non-functional requirements
  - Constraints on the services or functions offered by the system such as timing constraints, constraints on the development process, standards, etc.
  - Often apply to the system as a whole rather than individual features or services.
- Domain requirements
  - Constraints on the system from the domain of operation

Project Requirements

- Projects certainly do have other expectations and deliverables that are not a part of the software team implements, but that are necessary to the successful completion of the project as a whole.
- Project requirements include:
  - Physical resources the development team needs
  - Staff training needs
  - User documentation
  - Support documentation
  - Infrastructure changes needed in the operating environment
  - Product certification and compliance requirements
  - …

Functional Requirement

- Describe functionality or system services.
- Depend on the type of software, expected users and the type of system where the software is used.
- Functional user requirements may be high-level statements of what the system should do.
- Functional system requirements should describe the system services in detail.

Functional Requirement
Case Study: Mentcare system

- A user shall be able to search the appointments lists for all clinics.
- The system shall generate each day, for each clinic, a list of patients who are expected to attend appointments that day.
- Each staff member using the system shall be uniquely identified by his or her 8-digit employee number.

Requirements imprecision

- Problems arise when functional requirements are not precisely stated.
- Ambiguous requirements may be interpreted in different ways by developers and users.
- Consider the term ‘search’ in requirement 1
  - User intention – search for a patient name across all appointments in all clinics;
  - Developer interpretation – search for a patient name in an individual clinic. User chooses clinic then search.

Requirements completeness and consistency

In principle, requirements should be both complete and consistent.

Complete

- They should include descriptions of all facilities required.

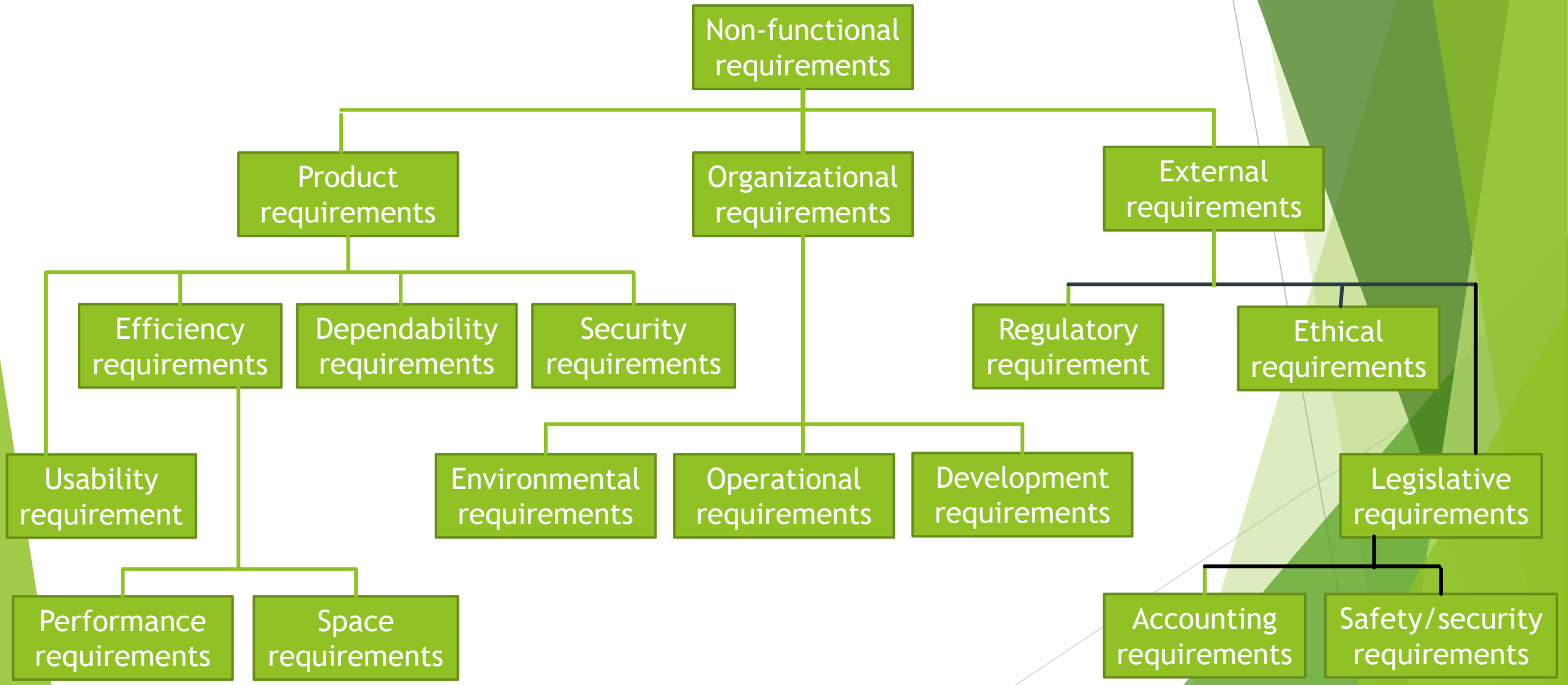
Consistent

- There should be no conflicts or contradictions in the descriptions of the system facilities.
- In practice, because of system and environmental complexity, it is impossible to produce a complete and consistent requirements document.

Non-functional Requirement

- These define system properties and constraints e.g. reliability, response time and storage requirements. Constraints are I/O device capability, system representations, etc.
- Process requirements may also be specified mandating a particular IDE, programming language or development method.
- Non-functional requirements may be more critical than functional requirements. If these are not met, the system may be useless.

Types of nonfunctional requirements



Non-functional requirements implementation

- Non-functional requirements may affect the overall architecture of a system rather than the individual components.
- For example, to ensure that performance requirements are met, you may have to organize the system to minimize communications between components.
- A single non-functional requirement, such as a security requirement, may generate a number of related functional requirements that define system services that are required.
- It may also generate requirements that restrict existing requirements.

Non-functional classifications

Product requirements

- Requirements which specify that the delivered product must behave in a particular way e.g. execution speed, reliability, etc.

Organizational requirements

- Requirements which are a consequence of organizational policies and procedures e.g. process standards used, implementation requirements, etc.

External requirements

- Requirements which arise from factors which are external to the system and its development process e.g. interoperability requirements, legislative requirements, etc.

Examples of nonfunctional requirements in the Mentcare system

- Product requirement

The Mentcare system shall be available to all clinics during normal working hours (Mon–Fri, 0830–17.30). Downtime within normal working hours shall not exceed five seconds in any one day.

- Organizational requirement

Users of the Mentcare system shall authenticate themselves using their health authority identity card.

- External requirement

The system shall implement patient privacy provisions as set out in HStan-03-2006-priv.

Goals and requirements

Non-functional requirements may be very difficult to state precisely and imprecise requirements may be difficult to verify.

Goal

- A general intention of the user such as ease of use.

Verifiable non-functional requirement

- A statement using some measure that can be objectively tested.

Goals are helpful to developers as they convey the intentions of the system users.

Usability requirements

- The system should be easy to use by medical staff and should be organized in such a way that user errors are minimized. (Goal)
- Medical staff shall be able to use all the system functions after four hours of training. After this training, the average number of errors made by experienced users shall not exceed two per hour of system use. (Testable non-functional requirement)

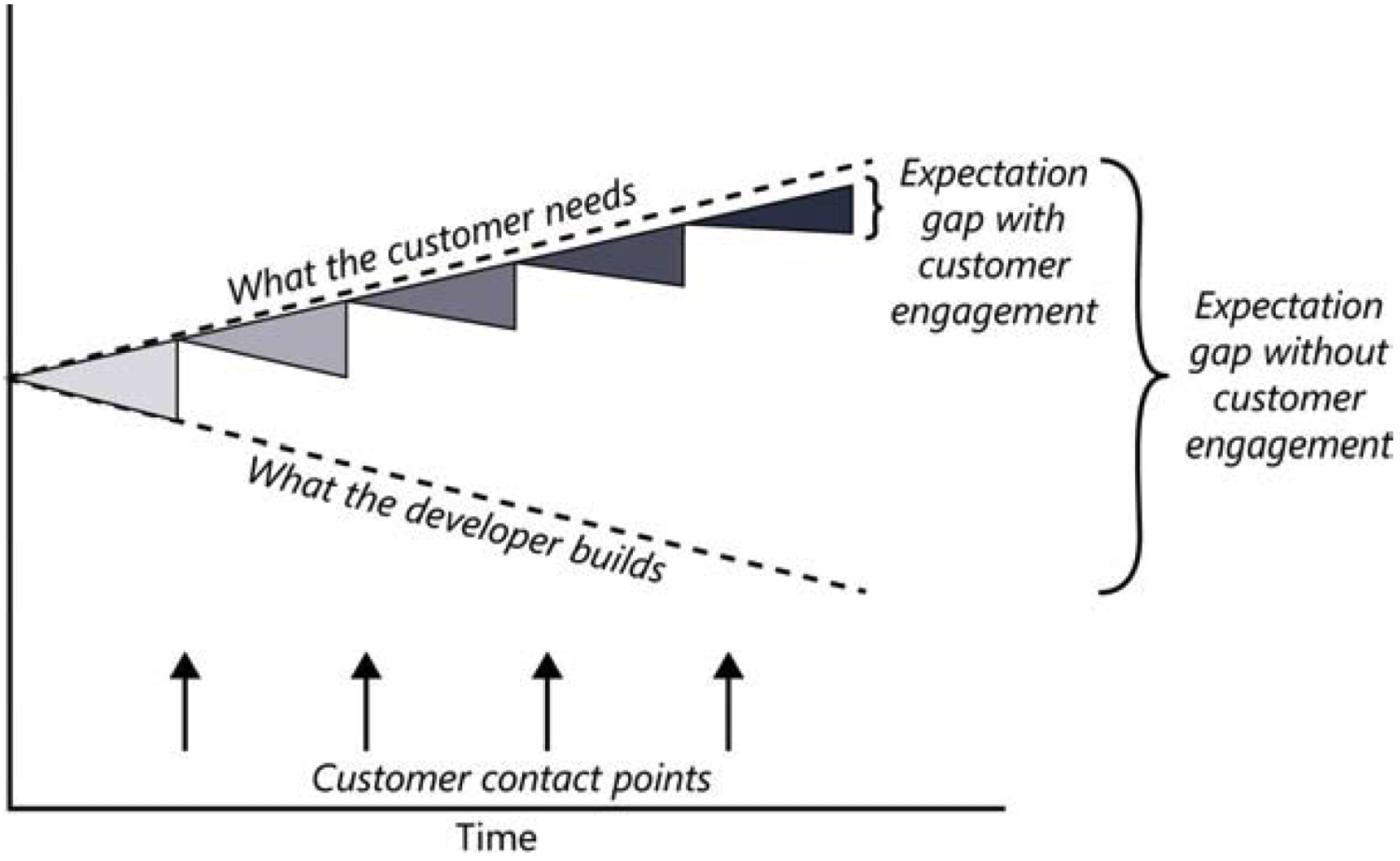
Metrics for specifying nonfunctional requirements

Property	Measure
Speed	Processed transactions/second User/event response time Screen refresh time
Size	Mbytes Number of ROM chips
Ease of use	Training time Number of help frames
Reliability	Mean time to failure Probability of unavailability Rate of failure occurrence Availability
Robustness	Time to restart after failure Percentage of events causing failure Probability of data corruption on failure
Portability	Percentage of target dependent statements Number of target systems

Traceability

- Traceability is a software engineering term that refers to documented links between software engineering work products.
- A traceability matrix allows a requirements engineer to represent the relationship between requirements and other software engineering work products.
- Traceability matrices often can be used to ensure the engineering work products have taken all requirements into account.

The Expectation Gap



In the next lecture…

- We will talk about requirements elicitation.

Question session